MODELX 2 / PROJECT SUMMARY

BloodLink

Connecting donor outreach with institution-led coordination

The challenge

Blood banks and hospitals need to coordinate urgent donor outreach while respecting donor choice and leaving suitability decisions with qualified staff. BloodLink provides shared tools for institution teams, network administrators and donors to manage that coordination in one connected workflow.

One request, three workspaces

INSTITUTION
Create and monitor a request

›

NETWORK ADMIN
Review and approve

›

DONOR APP
Review and choose to respond

What the system supports

DONOR MOBILE APP

- Profile, consent and availability
- Browse requests and submit a response
- Notifications, history and profile tools

INSTITUTION DASHBOARD

- Create and track requests
- Review stock and donor responses
- Manage team access and health alerts

NETWORK ADMIN

- Manage institution accounts
- Review request approvals
- Oversee the connected network

How outreach matching works

The current matching flow is a rule-based outreach filter using a donor's recorded blood-group label, consent, availability and location rules. It is not machine learning, a transfusion-compatibility calculation or medical eligibility assessment. Donors decide whether to respond; institution staff determine suitability and manage any next steps.

Implementation

The donor application uses React Native and TypeScript. The web dashboards use React and TypeScript. A Node.js/Express API connects the clients to PostgreSQL/PostGIS, with role-based access, request lifecycle controls, audit records and persisted in-app notifications. The documented local build supports the end-to-end request and response workflow; production hosting and external email or push delivery are not established by the local demo.

MODELX 2 / EVIDENCE & LIMITATIONS

What the current build demonstrates

Dataset and engineering evidence

The checked-in baseline snapshot is dated 3 November 2026 and is generated synthetic test data, marked SYNTHETIC_TEST_DATA_NOT_REAL. It contains fictional workflow records only; counts below describe test-fixture scale, not real donors, demand or inventory.

10,000 **simulated donors**	**48** **mock institutions**	**333** **requests**
274 **responses**	**384** **stock records**	**71,097** **history rows**

The baseline file validator reports 297,580 of 297,580 assertions passed. Project records also describe earlier import/rollback, authentication, API and interface checks using synthetic fixtures; those checks were not rerun for this summary. They provide software/data-validation evidence only, not evidence of health impact or real-world performance.

Real-data finding

No real-data analysis or result is available in this build. No licensed external dataset has been integrated, and BloodLink has not produced a real-data trend, risk, anomaly or forecast. The current video is a 51-second scripted walkthrough of synthetic local records, not the required three-minute real-data demonstration.

Data rights and next steps

The official Sri Lankan NBTS 2024 annual report was reviewed but not copied because its notice restricts unauthorized reproduction or modification. WHO's 2025 global report offers aggregate data through 2023 under CC BY-NC-SA 3.0 IGO terms, but it has not been integrated or analyzed. Any reuse must follow the license. Approximate town-centre coordinate provenance also remains to be documented. Source review: docs/MODELX_DATA_SOURCE_REVIEW.md.

- Select a dataset with documented permission, reporting period, retrieval date and known coverage limits.
- Build and validate an aggregate insight for a named population-level decision maker; show uncertainty and limits.
- Update the summary and report with reproducible real-data results, then record the required three-minute demonstration.
- Document the origin or independent authorship of map coordinates before relying on them.